

Basic Mathematics

Mathematics

Lecture Notes

A friendly path through algebra, geometry, and calculus.

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Chapter 1

Introduction

These notes are an invitation to look at mathematics in a different way. For many students, mathematics has long appeared as a collection of formulas, rules, and mnemonic patterns: one recognizes the type of problem, chooses a remembered procedure, performs the required operations, and moves on. Such skills have their place, but they are not the heart of mathematics. Mathematics begins when we stop for a moment and ask what we are really looking at, what kind of object has appeared, what question is being asked, what is known, what is unknown, and why a proposed step is justified.

The purpose of this course is therefore not only to practise calculations. We will learn how to read mathematical expressions, how to formulate precise questions, how to distinguish an object from its notation, how to notice structure, and how to build simple but meaningful chains of reasoning. A solved problem is not valuable only because it produces an answer. It is valuable because it can reveal a method of thought: how to begin, what to observe, which assumptions matter, what can be transformed, what remains unchanged, and how a conclusion follows from earlier facts.

The course will focus on three fundamental parts of mathematics: linear algebra, analytic geometry, and calculus. Linear algebra introduces vectors, matrices, systems of equations, and transformations. Analytic geometry shows how algebraic equations describe points, lines, planes, directions, distances, and angles. Calculus gives a language for functions, limits, continuity, derivatives, integrals, change, and accumulation. These topics are not isolated chapters placed next to one another. Together they form a basic language for describing structure, space, and change.

We will try to study this language from the inside. A definition should not be treated as a sentence to memorize, but as a way of introducing a new kind of object. A theorem should not be treated as a decorative statement, but as the result of asking a clear question under precise assumptions. An example should not be treated as a template to copy blindly, but as a small experiment in reasoning. Even a simple calculation can show something important if we ask why it works and what it tells us.

For this reason, some arguments will deliberately return in different places. The same habit of asking what is fixed and what is variable appears when we study functions, vectors, equations, and transformations. The same need for precision appears when we define a line, compute a derivative, solve a system of equations, or interpret an integral. The same mathematical question can often be seen algebraically, geometrically, and conceptually. This repetition is intentional: it helps the reader recognize patterns of reasoning rather than only patterns of exercises.

The number of solved tasks is not the main measure of progress. What matters is how the tasks are solved. Did we understand the question? Did we identify the object correctly? Did we choose notation consciously? Did we know why each step was allowed? Did we pause to interpret the result? Did the calculation teach us something about the idea behind it? These questions are part of the work. They are not an addition to mathematics; they are mathematics being done carefully.

Mathematics is useful in physics, engineering, computer science, data analysis, modelling, and many other fields, but its value is not only practical. It is also a way of seeing. It teaches precision, discipline, imagination, and the ability to move from simple observations to general conclusions. These notes are meant to show a piece of that beauty, strength, and richness. The goal is not merely to survive mathematical techniques, but to begin to appreciate mathematics

as an art of asking good questions and giving well-reasoned answers.

Chapter 2

Linear Algebra

This chapter introduces the basic objects and methods of linear algebra.

2.1 Vectors

This section will introduce vectors and their basic operations.

2.2 Matrices

A matrix is one of the first mathematical objects that looks very simple and very mechanical. It is a rectangular table of numbers. At first sight, this may suggest that matrix theory is only a matter of arranging numbers and following rules. This would be a serious misunderstanding. A matrix is useful because it allows us to store information, describe relations, encode systems of equations, and represent transformations. The same object can be read as a table, as a rule of calculation, as a linear transformation, or as a compact description of a structured problem.

In this section we will learn the basic operations on matrices: addition, subtraction, multiplication by a number, multiplication of two matrices, powers of a matrix, and simple operations on rows. The goal is not only to perform these operations correctly. The goal is to understand what kind of object is being manipulated and why the operation is defined in that particular way.

2.2.1 What is a matrix?

Before we calculate with matrices, we should learn to read them. A matrix has rows and columns. The first question is therefore not “What formula should I use?”, but “What is the size of the object in front of me?”

Definition

A matrix with m rows and n columns is called an $m \times n$ matrix. It has the form

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

The number a_{ij} is the entry in row i and column j .

The notation a_{ij} is not decoration. It tells us how to locate information. The first index tells us the row, and the second index tells us the column. Thus a_{23} means: second row, third column. This is a small convention, but conventions of this kind are what allow mathematics to be precise.

Example

Consider the matrix

$$A = \begin{pmatrix} 1 & 2 & 5 \\ 0 & -1 & 4 \end{pmatrix}.$$

This is a 2×3 matrix. It has two rows and three columns. The entry a_{12} is equal to 2, because it is in the first row and second column. The entry a_{23} is equal to 4, because it is in the second row and third column.

A row can be read as a horizontal list of numbers. A column can be read as a vertical list of numbers. Later, when matrices are used to describe systems of equations or transformations, this distinction will become important. For now it is enough to remember that a matrix is not only a pile of numbers: it is a pile of numbers with a shape.

2.2.2 Equality of matrices

Two matrices can be equal only if they have the same size. After that, equality is checked entry by entry.

Definition

Let $A = (a_{ij})$ and $B = (b_{ij})$ be matrices of the same size $m \times n$. We say that $A = B$ if

$$a_{ij} = b_{ij}$$

for every row index i and every column index j .

This definition is simple, but it teaches an important habit. When comparing mathematical objects, we first ask whether they are objects of the same type. A number can be equal to a number, a vector to a vector of the same dimension, and a matrix to a matrix of the same size. If the shapes do not match, the comparison is already suspicious.

2.2.3 Addition, subtraction, and multiplication by a scalar

Addition of matrices is defined entry by entry. This is natural when two matrices represent the same kind of table. If two tables have the same shape, then the entry in position (i, j) in the sum should come from adding the entries in position (i, j) in the two original tables.

Definition

Let $A = (a_{ij})$ and $B = (b_{ij})$ be matrices of the same size. Their sum is the matrix

$$A + B = (a_{ij} + b_{ij}).$$

Their difference is the matrix

$$A - B = (a_{ij} - b_{ij}).$$

If λ is a number, then

$$\lambda A = (\lambda a_{ij}).$$

The condition “of the same size” is essential. We cannot add a 2×3 matrix to a 3×2 matrix, because there is no consistent way to pair the entries position by position.

Example

Let

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & -1 \\ 2 & 1 \end{pmatrix}.$$

Then

$$A + B = \begin{pmatrix} 1+0 & 2+(-1) \\ 3+2 & 4+1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 5 & 5 \end{pmatrix}.$$

Similarly,

$$A - B = \begin{pmatrix} 1-0 & 2-(-1) \\ 3-2 & 4-1 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 1 & 3 \end{pmatrix},$$

and

$$2A = \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix}.$$

For a slightly longer expression, we calculate each part carefully:

$$3B = \begin{pmatrix} 0 & -3 \\ 6 & 3 \end{pmatrix}, \quad 2A = \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix},$$

so

$$3B - 2A = \begin{pmatrix} 0 & -3 \\ 6 & 3 \end{pmatrix} - \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix} = \begin{pmatrix} -2 & -7 \\ 0 & -5 \end{pmatrix}.$$

The important point is not only the final matrices. The important point is the method: we keep the shape, and we work position by position.

2.2.4 Matrix multiplication: row meets column

Matrix multiplication is different. It is not defined entry by entry. The reason is that multiplication is meant to combine rows of the first matrix with columns of the second matrix. This operation is less obvious than addition, but it is one of the main reasons matrices are powerful.

Definition

Let $A = (a_{ij})$ be an $m \times n$ matrix and let $B = (b_{ij})$ be an $n \times p$ matrix. The product AB is the $m \times p$ matrix $C = (c_{ij})$, where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

In words: the entry c_{ij} is obtained by multiplying row i of A by column j of B .

The sizes tell us whether multiplication is possible:

$$(m \times n)(n \times p) = m \times p.$$

The inner dimensions must match. The outer dimensions give the size of the answer.

Remark

A useful way to read matrix multiplication is this: each entry of the product is a dot product of one row and one column. Therefore matrix multiplication is not just a bigger version of ordinary multiplication. It is a rule for combining structured information.

Example

Using the same matrices

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & -1 \\ 2 & 1 \end{pmatrix},$$

we calculate

$$AB = \begin{pmatrix} 1 \cdot 0 + 2 \cdot 2 & 1 \cdot (-1) + 2 \cdot 1 \\ 3 \cdot 0 + 4 \cdot 2 & 3 \cdot (-1) + 4 \cdot 1 \end{pmatrix} = \begin{pmatrix} 4 & 1 \\ 8 & 1 \end{pmatrix}.$$

Every entry has a small story. The entry in the first row and first column comes from the first row of A and the first column of B :

$$(1, 2) \cdot (0, 2) = 1 \cdot 0 + 2 \cdot 2 = 4.$$

The entry in the second row and first column comes from

$$(3, 4) \cdot (0, 2) = 3 \cdot 0 + 4 \cdot 2 = 8.$$

This is the kind of calculation in which one should not rush. The entries of the answer are not found by matching positions as in addition. They are found by moving through rows and columns.

2.2.5 The order of multiplication matters

For numbers, multiplication is commutative: $ab = ba$. For matrices, this is usually false. Sometimes both products AB and BA are defined, but they may give different answers.

Example

For the matrices from the previous example, we already found

$$AB = \begin{pmatrix} 4 & 1 \\ 8 & 1 \end{pmatrix}.$$

Now calculate the opposite product:

$$BA = \begin{pmatrix} 0 & -1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} -3 & -4 \\ 5 & 8 \end{pmatrix}.$$

Thus

$$AB \neq BA.$$

The two products use the same two matrices, but they ask different row-column questions.

This example is not a small technical exception. It reveals a central feature of matrix algebra: order carries meaning. If matrices represent operations, then AB means that one operation is performed after another. Changing the order may change the result.

Theorem

For matrices of compatible sizes, the following laws hold:

$$\begin{aligned} A + B &= B + A, \\ (A + B) + C &= A + (B + C), \end{aligned}$$

$$(AB)C = A(BC),$$

$$A(B + C) = AB + AC, \quad (A + B)C = AC + BC.$$

However, in general,

$$AB \neq BA.$$

The last statement should be remembered carefully. It does not say that AB is never equal to BA . It says that equality cannot be assumed. If we want to claim that two matrices commute, we must check it or prove it from special structure.

2.2.6 Zero, identity, and diagonal matrices

Some matrices play special roles. The zero matrix has all entries equal to zero. The identity matrix is the matrix that does nothing under multiplication, in the same way that the number 1 does nothing under ordinary multiplication.

Definition

The $n \times n$ identity matrix is

$$I_n = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}.$$

If A has a compatible size, then

$$IA = A, \quad AI = A.$$

A diagonal matrix is a square matrix whose entries outside the main diagonal are zero.

Diagonal matrices are especially easy to understand. They act independently on each coordinate. This is why they are often the first matrices in which one can see structure instead of only arithmetic.

Example

Let

$$D = \text{diag}(2, -3, 5) = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

and let

$$E = \text{diag}(a, b, c) = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}.$$

Then

$$DE = \text{diag}(2a, -3b, 5c)$$

and

$$ED = \text{diag}(2a, -3b, 5c).$$

Therefore $DE = ED$. In this special case the matrices commute because every coordinate is scaled independently, and no coordinate is mixed with another one.

Powers of a diagonal matrix are also easy:

$$D^3 = \text{diag}(2^3, (-3)^3, 5^3) = \text{diag}(8, -27, 125).$$

If no diagonal entry is zero, then the inverse diagonal matrix is obtained by taking reciprocals:

$$D^{-1} = \text{diag}\left(\frac{1}{2}, -\frac{1}{3}, \frac{1}{5}\right).$$

A complete study of inverse matrices will come later. Here the diagonal case is useful because the idea is visible without complicated computation.

2.2.7 Rows, columns, and elementary row operations

Matrices are often manipulated row by row. A row operation changes the matrix, but it does so in a controlled and visible way. Such operations will later be used to solve systems of linear equations.

Example

Consider

$$C = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \\ 0 & 2 & -1 \end{pmatrix}.$$

First swap the first and third rows. We obtain

$$C_1 = \begin{pmatrix} 0 & 2 & -1 \\ -1 & 3 & 1 \\ 1 & 0 & 2 \end{pmatrix}.$$

Now add twice the new first row to the second row. The second row becomes

$$(-1, 3, 1) + 2(0, 2, -1) = (-1, 7, -1).$$

Therefore the final matrix is

$$C_2 = \begin{pmatrix} 0 & 2 & -1 \\ -1 & 7 & -1 \\ 1 & 0 & 2 \end{pmatrix}.$$

The point of writing the intermediate step is not to make the solution longer. It makes the reasoning inspectable. We can see exactly which row changed and why.

2.2.8 Column vectors as matrices

A vector can be written as a matrix with one column. This is not just a change of notation. It allows us to use matrix operations to organize vector calculations.

Let

$$u = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}, \quad v = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}.$$

Then

$$u + v = \begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix}, \quad u - v = \begin{pmatrix} -1 \\ -2 \\ 4 \end{pmatrix}.$$

The transpose of a column vector is a row vector:

$$v^\top = \begin{pmatrix} 2 & 0 & -1 \end{pmatrix}.$$

Thus the product $uv^\top$ is a 3×3 matrix:

$$uv^\top = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \begin{pmatrix} 2 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & -1 \\ -4 & 0 & 2 \\ 6 & 0 & -3 \end{pmatrix}.$$

Similarly,

$$vu^\top = \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix} \begin{pmatrix} 1 & -2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & -4 & 6 \\ 0 & 0 & 0 \\ -1 & 2 & -3 \end{pmatrix}.$$

These two matrices are not the same. Again, the order of multiplication matters.

There is also a structural observation hidden in this example. In the matrix $uv^\top$, every column is a multiple of u . Therefore all columns point in the same direction in the sense of linear dependence. This is why, if u and v are non-zero, the matrix $uv^\top$ has rank 1. We do not need the full theory of rank yet, but the example shows how a calculation can reveal structure.

2.2.9 Powers of a matrix

Powers of a matrix are defined only for square matrices. If P is square, then

$$P^2 = PP, \quad P^3 = PPP,$$

and so on. Matrix powers are useful when the same operation is repeated.

Example

Let

$$P = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}.$$

Then

$$P^2 = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 1 \end{pmatrix}$$

and

$$P^3 = \begin{pmatrix} 2 & 3 & 3 \\ 3 & 2 & 3 \\ 3 & 3 & 2 \end{pmatrix}.$$

The computation is ordinary matrix multiplication, but after calculating we should pause. The matrices show a cyclic pattern: the rows are related by rotation of positions. This is not an accident; it reflects the structure already present in P .

A good exercise is not finished when the arithmetic ends. It is finished when we have also asked what the arithmetic reveals.

2.2.10 A first glimpse: matrices as transformations

One reason matrices become powerful is that they can describe transformations. For example, the matrix

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

represents rotation of the plane by angle θ around the origin. This interpretation is not needed to multiply the matrices, but it helps explain why the product should have a meaningful form.

If we rotate by θ_2 and then by θ_1 , the total effect should be a rotation by $\theta_1 + \theta_2$. Algebra confirms this:

$$R(\theta_1)R(\theta_2) = \begin{pmatrix} \cos \theta_1 & -\sin \theta_1 \\ \sin \theta_1 & \cos \theta_1 \end{pmatrix} \begin{pmatrix} \cos \theta_2 & -\sin \theta_2 \\ \sin \theta_2 & \cos \theta_2 \end{pmatrix}.$$

The first entry of the product is

$$\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 = \cos(\theta_1 + \theta_2).$$

The other entries are obtained in the same way from trigonometric addition formulas, and the result is

$$R(\theta_1)R(\theta_2) = R(\theta_1 + \theta_2).$$

This is a beautiful moment: a matrix calculation, a geometric transformation, and a trigonometric identity all describe the same structure.

2.2.11 How to work with matrix problems

When working with matrices, do not begin by searching for a memorized rule. Begin by reading the object.

Summary

For matrix problems, ask the following questions.

- What is the size of each matrix?
- Are the operations even defined?
- If I add or subtract matrices, am I working entry by entry?
- If I multiply matrices, which row meets which column?
- Does the order of multiplication matter in this problem?
- Is there special structure: zero matrix, identity matrix, diagonal matrix, symmetry, repeated rows, or a visible pattern?
- After the computation, what did the result tell me?

The last question is essential. Matrix algebra is not only a technique for producing new arrays of numbers. It is a language for seeing structure. A student who learns to pause, read the objects, and interpret the result is already doing more than mechanical calculation.

2.3 Systems of Linear Equations

This section will introduce systems of linear equations and their connection with matrices.

Chapter 3

Analytic Geometry

This chapter introduces geometric objects described using coordinates and algebraic equations.

3.1 Coordinates

This section will introduce coordinate systems and the representation of points.

3.2 Lines

This section will introduce equations of lines and their geometric interpretation.

3.3 Planes

This section will introduce equations of planes in three-dimensional space.

Chapter 4

Calculus

This chapter introduces functions, limits, continuity, and derivatives.

4.1 Functions

This section will introduce functions as mathematical objects describing dependence between quantities.

4.2 Limits

This section will introduce the concept of a limit.

4.3 Continuity

This section will introduce continuity and its basic interpretation.

4.4 Derivatives

This section will introduce derivatives as rates of change.